

ALAN LAMB

Billings, MT | Available for remote contract work | (406) 598-4073 | alandaniellamb2@gmail.com | [linkedin.com/in/alanlamb-econ](https://www.linkedin.com/in/alanlamb-econ) | lambcast.net

TECHNICAL SKILLS

Quantitative Analysis & Reporting: Scenario analysis, program evaluation, regulatory impact analysis, benefit-cost analysis, causal inference

Econometrics: Regression modeling (OLS, fixed effects, DiD, synthetic control), panel data methods, wild cluster bootstrap, structural break testing, ARIMA forecasting, gradient-boosted tree models (XGBoost)

Software & Tools: Stata (advanced), R (intermediate), Python (intermediate), Excel, SQL, Git, Microsoft Office Suite

Data: EIA Form 930 API, LBNL interconnection queue data, multi-source merging (Census, BEA, NOAA), large dataset cleaning, validation, and reconciliation

EDUCATION

M.S. Applied Economics (STEM-designated) | University of Maryland | Expected December 2026

Coursework: Empirical Analysis I-III, Environmental Economics, Microeconomic Analysis, Economic Applications of R Programming

Master of Public Administration | University of Montana | May 2013

B.S. Political Science | Montana State University Billings | April 2010

APPLIED RESEARCH

AI Infrastructure and Regional Electricity Demand: Evidence from U.S. Interconnection Queues

Independent Research | March 2026 | SSRN: ssrn.com/abstract=6446446 | Dashboard: bit.ly/4siJ0VL | Replication Package: doi.org/10.7910/DVN/F8JV2T

- Built a scenario-based forecasting pipeline for regional electricity demand across ERCOT, PJM, and MISO: ARIMA baseline captured the historical time-series structure; queue-augmented models incorporated investment pipeline signal; and historical completion rates (ERCOT 33.9%, PJM 19.7%, MISO 17.8%) bounded conservative and trend scenarios through 2027
- Isolated AI data center investment as the source of minimum-demand divergence using a four-layer identification strategy (panel regression, synthetic control, difference-in-differences, narrative validation) where convergence across methods with different failure modes provided the evidence
- Headline result: 34.8 index-point divergence in ERCOT minimum hourly demand above a synthetic counterfactual; zero of three placebos exceeded the gap; robust across all ten panel specifications and three donor pool configurations
- Reconciled 184,000+ hourly EIA observations with 16,000+ LBNL interconnection queue filings from 2019 to 2025; maintained documented assumptions and quality-control checks so results were traceable and reproducible
- Produced an interactive published dashboard in Hex translating technical results into visuals for non-technical audiences, including zone-level load growth, queue filings by region, and forward demand projections

Disposable Bag Tax Program Evaluation: Difference-in-Differences Analysis

ECON 675 Environmental Economics | University of Maryland, Winter 2026 | Tools: Stata

- Evaluated behavioral response to Montgomery County's \$0.05 bag tax using a quasi-experimental design with 16,251 observations across DC, MD, and VA; structured the project as an applied causal-inference exercise on a real policy intervention with imperfect data
- Applied difference-in-differences methodology to estimate program effects on three margins: extensive margin (bag usage probability), intensive margin (bags per user), and total demand; controlled for store characteristics, demographics, time-of-day effects, and incentive programs to isolate the causal effect of the policy
- Quantified program impact: Maryland plastic bag use decreased 42 percentage points, reusable bag use increased 33 percentage points post-tax, and total plastic demand fell by 1 bag per shopper; documented the analysis end-to-end so the result was reproducible and defensible to a non-technical audience

EPA Uranium Mining Regulation: Benefit-Cost Analysis

ECON 675 Environmental Economics | University of Maryland, Fall 2025 | Grade: 96/100 | Tools: Stata

- Constructed a full welfare model evaluating proposed EPA in-situ uranium recovery regulations using EIA production and price data; estimated supply and demand elasticities from published literature and built baseline and post-policy equilibrium models for domestic, foreign, and aggregate markets
- Produced a benefit-cost tableau showing compliance costs, deadweight loss, and distributional welfare changes across stakeholder groups, using the same analytical framework applied in federal regulatory impact analysis and utility rate-case filings

The Missing Half of the Queue: Why FERC Should Require Load-Side Interconnection Queue Disclosure

Policy Brief | April 2026 | SSRN: ssrn.com/abstract=6614823

- Argued for a FERC Notice of Proposed Rulemaking requiring RTOs to publish large-load interconnection queue data in machine-readable form; traced the thirty-year regulatory history of generation-side transparency under Orders 888, 2000, 2003, and 2023
- Translated quantitative forecast uncertainty bounds into a written regulatory argument for a non-technical policy audience, the same translation work required for regulatory filings and stakeholder reporting

PROFESSIONAL EXPERIENCE

Montana Brewing Company & Hooligans Sports Bar | Billings, MT March 2010 - February 2024

Front of House Manager - June 2019 - February 2024

Bartender - October 2011 - June 2019

Security / Barback - March 2010 - October 2011

- Progressed from entry-level to managing FOH operations across a high-volume, dual-concept venue; directed teams of 20 to 40 per shift, oversaw hiring and onboarding, and handled escalations and service recovery across a 14-year tenure